RC Design

Project 01

We can delete this since it no need

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1 How to Design

- Building Load contains : **gravity** and **lateral** Loads.

1.1 Convert Area Loads to Line Load on Beams

Slabs distribute their load to beam depending on slab span direction and support condition (Eurocode 2, Section on Slabs [1]; IStructE Manual, Chapter 4 [6]).

The Tributary Width Method (IStructE Manual, Chapter 4 [6]):

For a **one-way slab**, load goes to the beams on the short sides:

Beam Line Load (kN/m) = Floor Load (kN/m2) $\times$ Tributary Width (m)

Tributary Width = half the slab span on each side of the beam (IStructE Manual, Cl. 3.2 [6]).

Example: If Slab spans 5m between two beams, so each beam take 5/2 = 2.5m tribuyary width. And if floor load is 10kN/m2, then beam line load is $10 \times 2.5 = 25$kN/m

For a **two-way slab**, load distributes in both directions (Eurocode 2, Section on Two-Way Slabs [1]; IStructE Manual, Chapter 5 [6]):

- Short span beam take triangular load
- Long span beam take trapezoidal load

1.2 Apply ULS and SLS Load Combinations

- ULS (Ultimate Limit State) - for stength design

$$\sum_{i} \gamma_{G,i} G_{k,i} + \gamma_{Q,1} Q_{k,1} + \sum_{j>1} \gamma_{Q,j} \varphi_{0.j} Q_{k,j}$$

- SLS (serviceability Limit State) - for deflection/crack check

  Chearacteristic Combinations:

$$\sum_{i} G_{k,i} + Q_{k,1} + \sum_{j>1} \varphi_{0,j} Q_{k,j} + (P_k)$$

Quasi-permanent (for long-term deflection)

$$\sum_i G_{k,i} + \sum_j \varphi_{2,j} Q_{k,j} + (P_k)$$

1.3 Beam Support Conditions

Support conditions determine the bending moment diagram shape and therefore where reinforcement must be placed (Eurocode 2, Section on Structural Analysis [1]; IStructE Manual, Chapter 4 [6]):

Table 1.1: Beam support condition

Support Type	Description	Moment at Support
Simply supported	Beam sits on wall/pad	Zero moment at ends
Continuous beam	Multiple spans, monolithic	Hogging moment at supports
Fixed end	Fully restrained (rare in RC)	Full fixity moment
Partially fixed	Column/beam junction	Partial moment transfer

Support Type	Rotation Allowed	Moment at Support	Typical Structure
Simply Supported	Yes	0	Beam on columns
Continuous	Small rotation	Negative moment	Multi-span floor beam
Fixed	No rotation	Maximum moment	Cantilever
Partially Fixed	Limited rotation	Partial moment	Beam-column joint

In reinforced concrete buildings, beams are almost always continuous over columns because they are cast monolithically (IStructE Manual, Chapter 4, Cl. 4.2 [6]).

1.4 Design Workflow

- **STEP 1:** Establish building layout and use
- **STEP 2:** Identify slab type (one-way / two-way)
- **STEP 3:** Gather G_k, permanent loads
  (EN 1991-1-1:2002, Annex A, Table A.1 [3])
- **STEP 4:** Gather Q_k, variable loads
  (EN 1991-1-1:2002, Cl. 6.3.1, Table 6.2 [3])
- **STEP 5:** Apply ULS combination

- **STEP 6:** Convert to beam line load
- **STEP 7:** Calculate bending moments and shear
  (IStructE Manual, Ch. 4 [6])
  (Eurocode 2, Structural Analysis [1])
- **STEP 8:** Size beam using L/d ratio
  (Eurocode 2, Deflection Section [1])
- **STEP 9:** Design reinforcement
  (Eurocode 2, ULS Design Section [1])
- **STEP 10:** Check SLS — deflection and cracking
  (EN 1990:2002, Cl. 6.5 [5])
  (Eurocode 2, SLS Section [1])
- **STEP 11:** Detail and produce drawings
  (Eurocode 2, Detailing Section [1])
  (IStructE Manual, Ch. 4 [6])

1.5 The Key Distinction: what are we Designing

According to (IStructE Manual, Chapter 3 [6]):

Element	How Load Works
Beam	Carries only load from its own floor
Column	Carries load from all floors above it
Foundation	Carries load from the entire building

Table 1.2: How structure work

1.6 The Unit Journey

Unit	Name	Used For	How We Get It
kN/m^3	Density	Material property	Given in [3] Table A.1
kN/m^2	Area load	Slab design	Density $\times$ thickness
kN/m	Line load	Beam design	kN/m^2 $\times$ tributary width
kN	Point force	Column/foundation design	kN/m $\times$ span $\div$ 2

Table 1.3: Unit of load to consider in design

2 Introduction

hello $y = ax + b = 2 + 3$ This is Our Team Intro.

3 Detailed Design Calculations

3.1 Loading

3.1.1 Live Load

For imposed laods on floor according to Table 6.1 – 6.4 in EN 1991-1-1:2002 [3]. Is summary in Table 3.1.

Table 3.1: Imposed Loads On Floor Base On EN 1991-1-1:2002

Category	**Sub-category / Use**	q_k **[kN/m²]**	Q_k **[kN]**
Category A	Areas for domestic and residential activities		
	- Floors	1,5 to 2,0	2,0 to 3,0
	- Stairs	2,0 to 4,0	2,0 to 4,0
	- Balconies	2,5 to 4,0	2,0 to 3,0
Category B	Office areas	2,0 to 3,0	1,5 to 4,5
Category C	Areas where people may congregate		
C1	Areas with tables (e.g. schools, cafes, restaurants, reading rooms)	2,0 to 3,0	3,0 to 4,0
C2	Areas with fixed seats (e.g. theatres, cinemas, churches)	3,0 to 4,0	2,5 to 7,0 (4,0)
C3	Areas without obstacles for moving people (e.g. museums, entrance halls)	3,0 to 5,0	4,0 to 7,0

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Table 3.1: Imposed Loads On Floor Base On EN 1991-1-1:2002

Category	Sub-category / Use	q_k [kN/m²]	Q_k [kN]
C4	Areas with possible physical activities (e.g. dance halls, gyms)	3,5 to 5,0	3,5 to 7,0
C5	Areas susceptible to large crowds (e.g. concert halls, sports halls with stands)	5,0 to 7,5	3,5 to 4,5
Category D	Shopping areas		
D1	Areas in general retail shops	4,0 to 5,0	3,5 to 7,0 (4,0)
D2	Areas in department stores	4,0 to 5,0	3,5 to 7,0
Category E	Storage and industrial activities (general storage, accessible areas)	7,5	7,0

And this is the imposed loads on second floor that we assumed. which shown in Table 3.2.

Table 3.2: Imposed Loads On Second Floor

No.	Room / Area	Category (EN 1991-1-1)	q_k (kN/m2)
1	Receptionist/ Reception	B	3
2	Workstation / Open office	B	3
3	Meeting Room	C1	3
4	President Office	B	3
5	Vice President Offices	B	3

Continued on next page

Table 3.2: Imposed Loads On Second Floor

No.	Room / Area	Category (EN 1991-1-1)	q_k (kN/m2)
6	Accounting	B	3
7	General Sale & Administration	B	3
8	Copy / Printer Area	B	3
9	Storages	E1	5
10	Lift Lobby / Lift Area	C3	4
11	Stair (Main Stair)	A	4
12	Restroom (Men)	B	3
13	Restroom (Women)	B	3
14	—	—	—
15	—	—	—
16	—	—	—
17	Entrance	C3	4
18	Emergency Stair	A	4
19	Server& Technical Room	E1 (conservative)	5
20	Kitchen / Pantry	C1	3
21	Library	C3	4
22	Share Space	C3	4

3.1.2 Dead Load

According to (EN 1990:2002, Cl. 4.1.1 [5]) Permanent loads are defined as loads whose variation in magnitude over time is negligible. The unit weights of construction materials used to calculate these loads are given in (EN 1991-1-1:2002, Annex A, Table A.1 [3]) and (EN 1991-1-1:2025, Annex A [2]):

Table 3.3: Dead Load Typical Value

Source	**Typical Value**
Reinforced concrete slab (200mm thick)	$0.2 \times 25 = 5.0\ \mathrm{kN/m^2}$
Screed/floor finish	$1.5\ \mathrm{kN/m^2}$
Ceiling + services (MEP)	$0.5\ \mathrm{kN/m^2}$
Partitions (treated as permanent)	$1,0\ \mathrm{kN/m^2}$
Total G_k	$8.0\ \mathrm{kN/m^2}$

References

[1] BSI, *BS EN 1992-1-1:2023 - Eurocode 2: Design of concrete structures - Part 1-1: General rules and rules for buildings, bridges and civil engineering structures*, BSI Standards Publication, 2023.

[2] BSI, *BS EN 1991-1-1:2025 - Eurocode 1: Actions on structures - Part 1-1: Specific weight of materials, self-weight of construction works and imposed loads for buildings*, BSI Standards Publication, 2025.

[3] CEN, *EN 1991-1-1:2002 - Eurocode 1: Actions on structures - Part 1-1: General actions - Densities, self-weight, imposed loads for buildings*, European Committee for Standardization, 2002. [Online]. Available: https://www.phd.eng.br/wp-content/uploads/2015/12/en.1991.1.1.2002.pdf

[4] CEN, *EN 1991-1-4:2005+A1 - Eurocode 1: Actions on structures - Part 1-4: General actions - Wind actions*, European Standard, Incorporating corrigendum January 2010, April 2010.

[5] CEN, *EN 1990:2002 - Eurocode: Basis of Structural Design*, European Committee for Standardization, 2002. [Online]. Available: https://www.phd.eng.br/wp-content/uploads/2015/12/en.1990.2002.pdf

[6] Institution of Structural Engineers (IStructE), *Manual for the Design of Concrete Building Structures to Eurocode 2*, 2nd ed., IStructE, London, 2006. [Online]. Available: https://archive.org/details/manual-for-the-design-of-concrete-building-structures-to-eurocode-2

Note that : We may use the referennces provide by professor (check in telegram group) if it still not enought we can add more maybe.

Appendices